Euclid's Elements — Book XI
Proposition 17

Formal Reconstruction — Technical Documentation

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If two straight lines be cut by parallel planes, they shall be cut in the same ratio.

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1 Introduction

Proposition XI.17 is the first proposition in Book XI where the spatial incidence theory and the classical theory of proportion interact in an essential way. Three parallel planes cut two transversals in the orders

$$A - E - B, \qquad C - F - D,$$

and Euclid proves

$$AE : EB = CF : FD.$$

The classical construction introduces a point O on AD in the middle plane. Proposition XI.16 then produces two pairs of parallel section lines, and Proposition VI.2 is applied in two different planar triangles:

$$AE : EB = AO : OD, \qquad AO : OD = CF : FD.$$

Proposition V.11 gives the final equality of ratios.

The formal reconstruction keeps this classical architecture, but it makes three hidden issues explicit.

First, the point O is not assumed: it is constructed synthetically from the spatial configuration by XI.16 and an ambient Pasch argument. Second, the two applications of VI.2 occur in two explicit induced plane geometries, not in the ambient 3-space. Third, segment proportion is represented by a purely synthetic raw relation based on right-triangle witnesses and angle congruence; no numerical length, segment division, multiplication, real coordinates, or quotient segment arithmetic is used.

One limitation remains visible in the production theorem: planar VI.2 is supplied as an explicit hypothesis

```
hVI2 :
  forall tau : S.Plane,
    HilbertVI2Raw (PlaneGeo Geo tau)
```

rather than derived here from the primitive Hilbert axioms. Thus XI.17 closes the spatial reconstruction around VI.2 without pretending that the whole planar proportion theory has already been reconstructed from first principles.

2 Classical Configuration

Let π_0, π_1, π_2 be three parallel planes. Two transversals meet them at

$$A, E, B$$

and

$$C, F, D,$$

respectively, with

$$A - E - B, \qquad C - F - D.$$

Thus

$$A, C \in \pi_0, \quad E, F \in \pi_1, \quad B, D \in \pi_2.$$

The conclusion is

$$AE : EB = CF : FD.$$

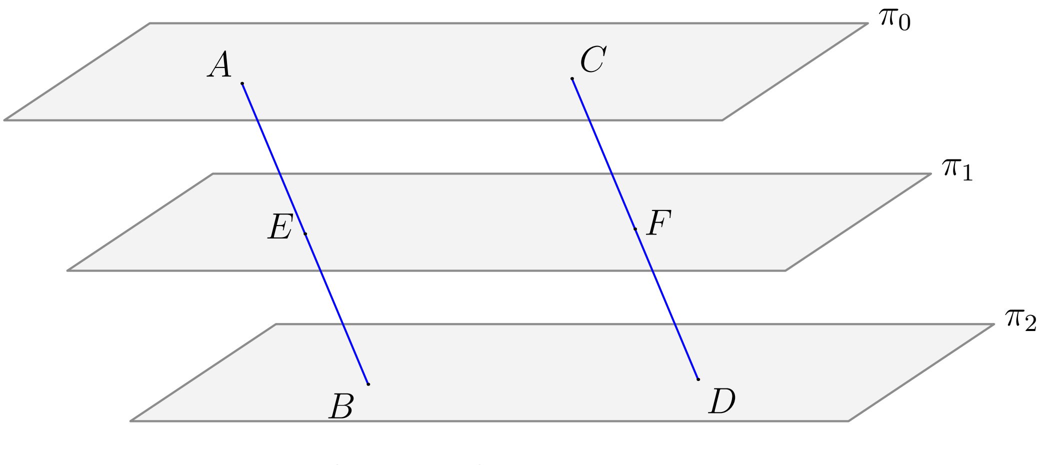

Figure 1: The classical input configuration of Proposition XI.17. The three light planes π_0, π_1, π_2 are pairwise parallel. The two genuinely spatial transversals are shown in blue and meet the planes in the orders $A - E - B$ and $C - F - D$.

The formal statement records pairwise parallelism explicitly:

$$\pi_0 \parallel \pi_1, \qquad \pi_1 \parallel \pi_2, \qquad \pi_0 \parallel \pi_2.$$

It also records the two betweenness relations directly.

3 Statement

Theorem 1 (Euclid XI.17)**.** *If two straight lines are cut by parallel planes, they are cut in the same ratio.*

In the notation of Figure 1,

$$AE : EB = CF : FD.$$

The current production declaration is:

```
theorem euclid_proposition_11_17
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (hVI2 :
      forall tau : S.Plane,
        HilbertVI2Raw (PlaneGeo Geo tau))
    (pi0 pi1 pi2 : S.Plane)
    (A E B C F D : Geo.Point)
    (hParallel01 :
      HilbertSpacePlanesParallelIncidence Geo pi0 pi1)
    (hParallel12 :
      HilbertSpacePlanesParallelIncidence Geo pi1 pi2)
    (hParallel02 :
      HilbertSpacePlanesParallelIncidence Geo pi0 pi2)
    (hApi0 : S.OnPlane A pi0)
    (hCpi0 : S.OnPlane C pi0)
    (hEpi1 : S.OnPlane E pi1)
    (hFpi1 : S.OnPlane F pi1)
    (hBpi2 : S.OnPlane B pi2)
    (hDpi2 : S.OnPlane D pi2)
    (hAEB : Geo.Between A E B)
    (hCFD : Geo.Between C F D)
    (hBD : Ne B D)
    (hAC : Ne A C) :
    HilbertSpaceSegmentProportionRaw
      Geo
      A E
      E B
      C F
      F D
```

The hypotheses hBD and hAC are explicit nondegeneracy conditions. They guarantee that the two outer-plane connector lines BD and AC used in Euclid's construction are genuine lines determined by distinct points.

4 Classical Proof

Join AC, BD, and AD. Let AD meet the middle plane π_1 at O, and join EO and OF.

The plane through E, B, D, O cuts the two parallel planes π_1, π_2. By XI.16 their common sections are parallel:

$$EO \parallel BD.$$

Similarly, the plane through A, O, F, C cuts the two parallel planes π_0, π_1, so XI.16 gives

$$AC \parallel OF.$$

Now work first in triangle ABD. Since E lies on AB, O lies on AD, and

$$EO \parallel BD,$$

VI.2 gives

$$AE : EB = AO : OD.$$

Next work in triangle ADC. Since O lies on AD, F lies on DC, and

$$OF \parallel AC,$$

VI.2 gives

$$AO : OD = CF : FD.$$

By V.11,

$$AE : EB = CF : FD.$$

This is exactly the classical proof pattern recorded in the standard text of XI.17: XI.16 is used twice, VI.2 is used twice, and V.11 composes the two proportions.

Diagrammatic audit. The first figure records the input state before Euclid's construction. The second records the stable proof state after the auxiliary point O and the two section-parallelisms have been obtained. No separate figure is needed for the raw ratio witnesses: they belong to the formal representation of proportion, not to a new Euclidean configuration.

5 The Auxiliary Point O

In Euclid's text the phrase "let AD meet the plane π_1 at O" is geometrically natural but formally nontrivial. The production theorem does not take O as input. It constructs O in

```
euclid_proposition_11_17_construct_O
```

with conclusion

```
exists O : Geo.Point,
  Geo.Between A O D /\
  S.OnPlane O pi1
```

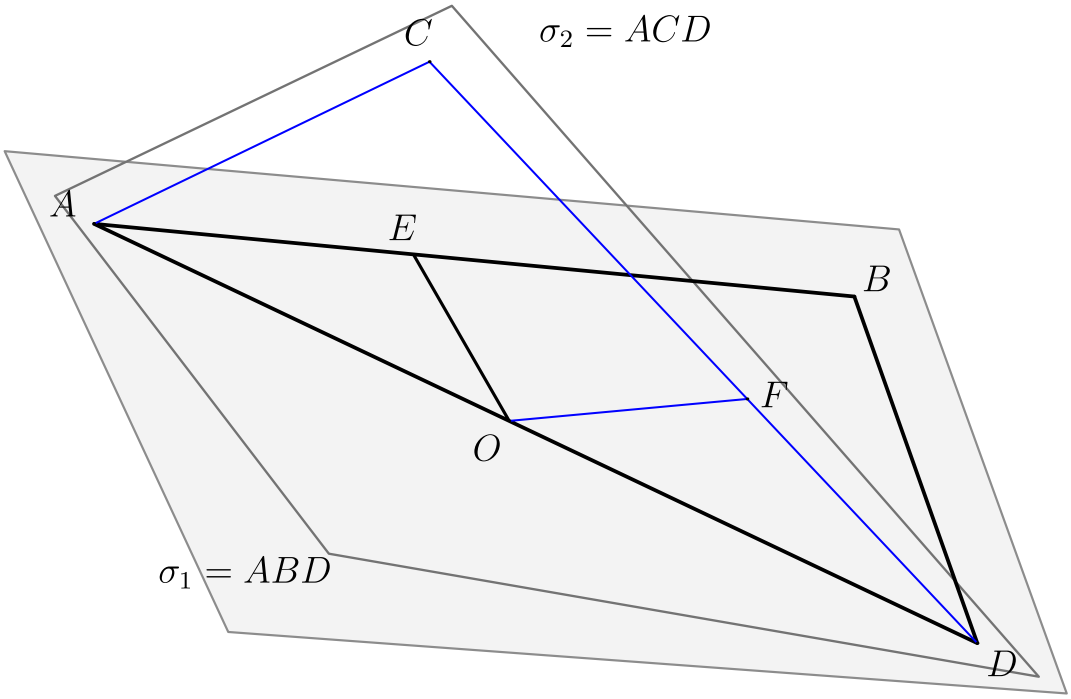

Figure 2: Euclid's auxiliary construction. The reference plane $\sigma_1 = ABD$ is shown as the main light plane: $E \in AB$, $O \in AD$, and $EO \parallel BD$. The second plane $\sigma_2 = ACD$ meets σ_1 along AD; the genuinely spatial part leaving the reference plane is shown in blue, with $F \in CD$ and $OF \parallel AC$. The two VI.2 applications therefore occur in different planar slices sharing the line AD.

The construction is synthetic.

First, the proof constructs the plane

$$\sigma = \mathrm{plane}(A, B, D).$$

Because $A - E - B$, the point E also lies in σ.

Next XI.16 is applied to the parallel planes π_1 and π_2, cut by σ. It produces the section line through E and the section line BD, and proves these two lines parallel. Hence the section through E is disjoint from BD.

The line through E in the middle-plane section meets the open side AB of triangle ABD. Ambient Pasch in σ says that it must exit through AD or BD. The BD alternative is impossible because the section line is parallel to BD. Therefore it meets AD at a point O, and the section characterization supplied by XI.16 shows

$$O \in \pi_1.$$

This is one of the main formal gains of the reconstruction: a diagrammatic intersection that Euclid takes as immediate is factored into XI.16 plus the order-theoretic Pasch mechanism.

6 What is genuinely three-dimensional here?

XI.17 has two distinct spatial layers.

The first is the original arrangement of three parallel planes and two transversals. The point O cannot be constructed inside one fixed pre-existing plane; its existence depends on the interaction between the middle plane and the auxiliary plane ABD.

The second is the construction of the two section-parallelisms

$$EO \parallel BD, \qquad AC \parallel OF.$$

Each is produced by XI.16 from a pair of ambient parallel planes and a cutting plane.

Only after these spatial steps are complete does the proof reduce to two planar slices. The first VI.2 calculation is performed in a plane τ_1 containing EO and BD; the second is performed in a plane τ_2 containing AC and OF. These need not be the same plane.

Thus the formal proof does not "flatten" the whole XI.17 configuration. It moves from ambient 3-space to two different local planar geometries and then transports the resulting proportions back to space.

7 Synthetic Representation of Ratio and Proportion

The production proof deliberately avoids segment arithmetic.

A concrete ratio $PQ : RS$ is represented by a right-triangle witness: one leg is congruent to PQ, the other to RS, and the corresponding acute angle represents the ratio. Equality of ratios is then expressed by congruence of the defining acute angles.

The planar relation is

```
HilbertSegmentProportionRaw
```

and its ambient 3D counterpart is

```
HilbertSpaceSegmentProportionRaw
```

The raw VI.2 interface is

```
def HilbertVI2Raw
    [HilbertIncidence Geo]
    [HilbertCongruence Geo] : Prop :=
  forall X E Y F Z : Geo.Point,
    Geo.Between X E Y ->
    Geo.Between X F Z ->
    Geo.Parallel E F Y Z ->
    HilbertSegmentProportionRaw
        Geo X E E Y X F F Z /\
      HilbertSegmentProportionRaw
        Geo E Y X E F Z X F
```

This is exactly the proportional consequence needed from VI.2, stated without division or numerical lengths.

The final V.11-type composition is supplied by transitivity of the raw proportion relation.

8 Spatial / Planar Architecture Used

The ambient layer of XI.17 uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpacePlanesParallelIncidence
HilbertSpaceLinesParallel
```

The key spatial theorem is

```
euclid_proposition_11_16
```

and the auxiliary point construction also uses ambient Pasch through

```
HilbertSpaceOrder.pasch_in_plane
```

The two VI.2 applications live in explicit induced plane geometries

```
PlaneGeo Geo tau1
PlaneGeo Geo tau2
```

The resulting planar proportions are transported back to ambient space by

```
hilbertSegmentProportionRaw_to_space
```

and finally composed by

```
hilbertSpaceSegmentProportionRaw_trans
```

This respects the permanent Book XI instance discipline: there is no global ambient planar order or planar congruence instance. Planar proportion arguments occur only after an explicit plane slice has been selected.

9 Proof Architecture of the Reconstruction

The production proof is organized into five mathematical stages.

9.1 Stage 1: construct O

The theorem

```
euclid_proposition_11_17_construct_O
```

constructs

$$A - O - D, \qquad O \in \pi_1.$$

As explained above, this uses the plane ABD, XI.16, and Pasch.

9.2 Stage 2: obtain the two XI.16 parallelisms

The spatial core constructs four lines:

$$EO, \quad BD, \quad AC, \quad OF$$

and proves

$$EO \parallel BD, \qquad AC \parallel OF.$$

The corresponding production helper is

```
euclid_proposition_11_17_spatial_core
```

This stage is purely about ambient incidence, order, and the section theorem XI.16.

9.3 Stage 3: first VI.2 slice

From the spatial parallelism $EO \parallel BD$, the proof opens a witness plane τ_1 containing both lines. The points A, E, B, O, D are placed in the induced geometry PlaneGeo(τ_1).

The local VI.2 interface then yields

$$AE : EB = AO : OD.$$

This planar proportion is transported back to ambient 3-space.

9.4 Stage 4: second VI.2 slice

The parallelism $AC \parallel OF$ similarly provides a plane τ_2. Inside PlaneGeo(τ_2), the proof uses the reversed betweenness orientations

$$D - O - A, \qquad D - F - C$$

to fit the exact argument order of the raw VI.2 interface.

After applying VI.2, the formal proportion is reversed back to the desired orientation by

```
hilbertSegmentProportionRaw_reverse_all
```

and transported to ambient space. The result is

$$AO : OD = CF : FD.$$

9.5 Stage 5: V.11

The two ambient raw proportions

$$AE : EB = AO : OD, \qquad AO : OD = CF : FD$$

are composed by

```
hilbertSpaceSegmentProportionRaw_trans
```

to obtain

$$AE : EB = CF : FD.$$

10 Wyler-Style Flat Reconstruction

The independent Wyler reconstruction of XI.17 now has a genuinely affine endpoint. Its spatial layer is expressed by flat/carrier calculus, while its final proportional statement is no longer obtained by crossing into `HilbertVI2Raw`. Instead, the Wyler path introduces its own affine notion of equality of division positions on transversals.

The first generated slice is

$$\sigma_1 = \text{plane}(A, B, D).$$

Because $A - E - B$ and the auxiliary point O satisfy $A - O - D$, both E and O lie in σ_1. Wyler XI.16 gives the exact section identities

$$C(\pi_1) \cap C(\sigma_1) = C(l_{EO}), \qquad C(\pi_2) \cap C(\sigma_1) = C(l_{BD}),$$

together with

$$l_{EO} \parallel l_{BD}.$$

The incidences $O \in l_{EO}$ and $D \in l_{BD}$ are recovered by direct rewriting with these carrier equalities.

The second generated slice is

$$\sigma_2 = \text{plane}(A, C, D).$$

Likewise,

$$C(\pi_0) \cap C(\sigma_2) = C(l_{AC}), \qquad C(\pi_1) \cap C(\sigma_2) = C(l_{OF}),$$

and XI.16 yields

$$l_{AC} \parallel l_{OF}.$$

Again, $C \in l_{AC}$ and $F \in l_{OF}$ follow from the meet identities.

This spatial skeleton is packaged by

```
euclid_XI17_wyler_first_sections
euclid_XI17_wyler_sections
```

The auxiliary point O is constructed in the same carrier-first spirit. In the plane

$$\sigma = \text{plane}(A, B, D),$$

the middle plane π_1 cuts σ in a line l_E, while π_2 cuts it in the line l_{BD}. Wyler XI.16 gives

$$l_E \parallel l_{BD}.$$

Since l_E meets the open side AB at E, Pasch forces it to meet AD or BD. The second branch contradicts the disjointness encoded by line parallelism. Hence there is O with

$$A - O - D, \qquad O \in \pi_1.$$

This is implemented by

```
euclid_proposition_11_17_construct_O_wyler
```

10.1 Affine division in the Wyler path

The key additional module is

```
CGJteamLab.Wyler.HilbertWylerProportion
```

It introduces the elementary affine intercept configuration

```
HilbertWylerDivisionStep
```

with geometric data

$$X - E - Y, \qquad X - F - Z, \qquad EF \parallel YZ.$$

This configuration expresses that E and F occupy the same affine division position on the two transversals. No numerical lengths, segment quotients, field operations, coordinates, or angle ratios occur in this definition.

The generated relation

```
HilbertWylerSameDivision
```

acts on ordered triples of points. Informally,

$$(A, E, B) \sim (C, F, D)$$

means that E divides AB in the same affine position in which F divides CD. The relation contains the elementary parallel-section step and is closed under reflexivity, symmetry, transitivity, and simultaneous reversal of the two ordered triples.

The first section pair gives directly

$$(A, E, B) \sim (A, O, D),$$

because

$$A - E - B, \qquad A - O - D, \qquad EO \parallel BD.$$

For the second pair the proof reverses the two transversals:

$$D - O - A, \qquad D - F - C, \qquad OF \parallel AC.$$

Hence

$$(D, O, A) \sim (D, F, C).$$

Simultaneous reversal gives

$$(A, O, D) \sim (C, F, D),$$

and transitivity yields the affine XI.17 conclusion

$$(A, E, B) \sim (C, F, D).$$

The production theorem is

```
euclid_proposition_11_17_wyler_affine
```

and its conclusion is

```
HilbertWylerSameDivision Geo A E B C F D
```

This theorem is the genuine Wyler endpoint of XI.17. It does not assume `HilbertVI2Raw`, does not use `HilbertSpaceCongruence`, and does not represent ratios by congruent acute angles. The older theorem

```
euclid_proposition_11_17_wyler
```

remains available as a compatibility result whose target is the previously used synthetic relation

```
HilbertSpaceSegmentProportionRaw
```

Thus the project retains the connection with the canonical Book V/VI proportion API without making that API foundational for the Wyler proof.

The complete Wyler dependency pattern is now

```
pi0 || pi1 || pi2
   |
   +--> sigma1 = plane(A,B,D)
   |       |
   |       +--> C(pi1) inter C(sigma1) = C(lEO)
   |       +--> C(pi2) inter C(sigma1) = C(lBD)
   |       +--> lEO || lBD
   |
   +--> sigma2 = plane(A,C,D)
   |       |
   |       +--> C(pi0) inter C(sigma2) = C(lAC)
   |       +--> C(pi1) inter C(sigma2) = C(lOF)
   |       +--> lAC || lOF
   |
   +--> Pasch in plane(A,B,D)
   |       |
   |       +--> O on AD and O in pi1
   |
   +--> HilbertWylerDivisionStep
```

```
          |
          +--> (A,E,B) ~ (A,O,D)
          +--> (A,O,D) ~ (C,F,D)
          |
          transitivity
          |
          (A,E,B) ~ (C,F,D)
```

The characteristic Wyler discipline is therefore complete in XI.17: first determine the generated flats and their exact meets, then express the remaining proportional content as an affine invariant of parallel sections. There is no remaining VI.2 boundary inside the affine theorem itself.

11 Lean Formalization

The public theorem is intentionally short at the top level. It first calls

```
euclid_proposition_11_17_construct_O
```

to obtain O, then delegates the complete remaining argument to

```
euclid_proposition_11_17_with_O
```

The latter handles two conceptually different tasks:

- reconstruction of Euclid's section lines and parallelisms;
- the two plane-local VI.2 calculations and their transport back to ambient space.

This separation is mathematically useful. It isolates the genuinely spatial existence problem for O from the proportion calculation that begins once O is available.

12 Hidden Formal Data

Several pieces of information visible in Euclid's diagram become explicit proof obligations.

- Pairwise parallelism of the three planes is given explicitly.
- The order relations $A - E - B$ and $C - F - D$ are hypotheses, not inferred from a picture.
- The connector points A, C and B, D are required to be distinct: $A \neq C$ and $B \neq D$.
- The point O must be proved to exist, to satisfy $A - O - D$, and to lie in the middle plane.
- The planes ABD and ACD must be constructed from verified noncollinearity.
- XI.16 supplies section lines pointwise; their incidences with E, O, B, D, A, C, F must be recovered from those characterizations.

- Each spatial line-parallelism carries a witness plane, and that plane becomes the correct `PlaneGeo` slice for VI.2.
- The second VI.2 application requires orientation normalization before the resulting proportion can be put into Euclid's final order.

These obligations explain why the formal DAG is longer than Euclid's visible proof even though the mathematical strategy is the same.

13 Relationship with Earlier Book XI Propositions

XI.16 is the decisive Book XI dependency, both classically and formally.

Classically it is invoked twice:

$$EO \parallel BD, \qquad AC \parallel OF.$$

Formally it plays an additional role in the construction of O: the middle and lower parallel planes are cut by the plane ABD, and the resulting section line through E is shown parallel to BD. This disjointness is what eliminates one of the two Pasch exits.

Thus the formal use of XI.16 is slightly richer than the visible textbook proof:

- once to construct O;
- twice to obtain the parallel section pairs used in the ratio proof.

No Wyler flat calculus is imported. XI.17 depends on the canonical synthetic XI.16 in the main Book XI route.

14 Relationship with Books V and VI / Hilbert Infrastructure

The classical proportion dependencies are exactly:

$$\text{VI.2}, \qquad \text{V.11}.$$

In the current formal architecture they appear in synthetic raw form.

VI.2 is represented by the explicit plane-local assumption

```
forall tau : S.Plane,
  HilbertVI2Raw (PlaneGeo Geo tau)
```

and V.11 by transitivity of

```
HilbertSpaceSegmentProportionRaw
```

The right-triangle representation of ratio uses Hilbert congruence and right-angle infrastructure. The production audit also confirms that no new XI.17-specific geometric axiom was introduced.

However, because VI.2 is an explicit hypothesis, the theorem should not yet be described as a derivation of Euclidean proportion theory from primitive Hilbert incidence, order, and congruence alone.

For the canonical theorem and the compatibility theorem `euclid_proposition_11_17_wyler`, this Book V/VI layer remains present. The new affine theorem `euclid_proposition_11_17_wyler_affine` bypasses it entirely: its conclusion is `HilbertWylerSameDivision`, obtained from parallel-section steps, reversal, and transitivity.

No Group IV typeclass appears in the theorem. The spatial section geometry itself remains independent of the Euclidean parallel postulate.

15 Dependency Summary

Classical.

$$\begin{gathered}
\pi_0 \parallel \pi_1 \parallel \pi_2, \quad A - E - B, \quad C - F - D \\
\downarrow \\
\text{join } AC, BD, AD; \quad O = AD \cap \pi_1 \\
\downarrow \\
\text{XI.16}: \; EO \parallel BD, \quad AC \parallel OF \\
\downarrow \\
\text{VI.2 in } \triangle ABD : AE : EB = AO : OD \\
\downarrow \\
\text{VI.2 in } \triangle ADC : AO : OD = CF : FD \\
\downarrow \\
\text{V.11} \\
\downarrow \\
AE : EB = CF : FD.
\end{gathered}$$

Formal.

```
Hilbert 3D incidence/order/congruence
  + HilbertVI2Raw on every PlaneGeo
        |
        v
Proposition11_16 + ambient Pasch
        |
        v
euclid_proposition_11_17_construct_O
        |
        v
euclid_proposition_11_17_spatial_core
        |
        v
PlaneGeo tau1: AE:EB = AO:OD
        |
        v
PlaneGeo tau2: AO:OD = CF:FD
        |
        v
transport to ambient raw proportions
        |
        v
hilbertSpaceSegmentProportionRaw_trans
        |
```

```
      v
euclid_proposition_11_17
```

The main conceptual difference is the treatment of O. Euclid names the intersection directly. Lean proves its existence from the spatial incidence/order infrastructure before the classical VI.2 chain can begin.

16 Audit Status

The canonical theorem `euclid_proposition_11_17` and the compatibility Wyler theorem `euclid_proposition_11_17_wyler` retain the audit profile

$$[\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Geometry.bookZero\_nullSegment1},\ \texttt{Quot.sound}].$$

This is consistent with their use of the synthetic raw proportion infrastructure.

The new affine theorem `euclid_proposition_11_17_wyler_affine` has the strictly smaller audit profile

$$[\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}].$$

In particular,

```
Geometry.bookZero_nullSegment1
```

disappears from the dependency set. This is exactly what the architecture predicts: the affine theorem no longer passes through the right-triangle/raw ratio machinery of `Hilbert3DProportion`.

The result therefore has two independent confirmations. First, its theorem signature contains no `HilbertVI2Raw` or `HilbertSpaceCongruence` hypothesis. Second, its axiom audit loses the null-segment axiom that remains visible in the synthetic proportion route. The Wyler reconstruction of XI.17 is consequently not merely a different organization of the same ratio proof; it has a genuinely affine formal endpoint.

The two figures continue to represent the stable geometric states used by both routes: the original three-plane transversal configuration and the post-construction state containing O and the two section-parallelisms.

17 Conclusion

Proposition XI.17 is the point where the Book XI reconstruction begins to combine genuinely spatial geometry with a statement that is classically expressed as a proportion.

The canonical route follows Euclid closely: XI.16 supplies the two section parallelisms, Pasch constructs the auxiliary point O, and two plane-local applications of VI.2 are composed by a V.11-type transitivity theorem. This route ends in `HilbertSpaceSegmentProportionRaw`.

The Wyler reconstruction now separates the same mathematics more sharply. Its three-dimensional content is expressed entirely through generated planes, exact carrier meets, parallel section lines, and Pasch. Its final invariant is the affine relation `HilbertWylerSameDivision`. The elementary parallel-section configuration gives equality of division position directly, and the proof of XI.17 consists of two such steps joined through the auxiliary transversal AD.

Thus XI.17 no longer contains the previously identified foundational gap in the Wyler path. The affine theorem does not assume VI.2 and does not use metric segment-ratio machinery. The older raw-proportion theorem remains only as a compatibility interface with the canonical Book V/VI development.

This gives XI.17 a particularly useful architectural role in the project: it shows that the flat/carrier calculus is strong enough not only to reconstruct the spatial configuration but also to carry the natural affine invariant associated with three parallel planes and their transversals.